

GENERAL O. OZOCHIAWAEZE

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EDUCATION

Purdue University Ph.D. in Mathematics (Inverse Problems, PDEs, Scattering) Advisors: Isaac Harris & Peijun Li	Aug 2021 – May 2027
New Jersey Institute of Technology M.S. in Applied Mathematics	May 2021
Rutgers University B.A. in Mathematics	May 2018

PUBLICATIONS & PREPRINTS

- 1 I. Harris and G. Ozochiawaeze, “Novel implementation of the extended sampling method for inverse biharmonic scattering” (working title), In preparation.
- 2 R. Ceja Ayala, I. Harris, and G. Ozochiawaeze, “Factorization method for the biharmonic scattering problem for an absorbing penetrable scatterer,” arXiv preprint arXiv:2511.05711
- 3 I. Harris, P. Li, and G. Ozochiawaeze, “Sampling methods for the inverse cavity scattering problem of biharmonic waves,” *Inverse Problems*, 42 (2026), 015002

PROFESSIONAL EXPERIENCE

Purdue University Graduate Researcher (Inverse Problems & Wave Imaging) • Advanced non-iterative imaging algorithms to reconstruct hidden structures from wave-based measurements, achieving reliable performance with limited-aperture data.	2023 – Present
MIT Lincoln Laboratory Signal Processing Research Intern (Radar & Acoustics) • Engineered sparse-data SAR radar reconstruction methods using compressive sensing and analyzed sonar localization under environmental noise and coherence loss.	2022 – 2023
New Jersey Institute of Technology Research Assistant • Formulated and simulated acoustic wave propagation in heterogeneous media via finite element and domain decomposition methods, examining sensitivity to high-frequency scattering.	2020 – 2021

TEACHING & SERVICE

Lecturer, Purdue University MA 16020 — Applied Calculus II	2025
Teaching Assistant, Purdue University STAT 416; MA 251; MA 161; MA 162	2021-2024
Mentoring & Service Directed Reading Program Mentor (2025–2026); Graduate Representative (2025–2026); Founder of Math History Seminar (2022–2025)	

AWARDS & HONORS

Bilsland Dissertation Fellowship (2026–2027) GEM Fellowship (2022–2023) Graduate C/Startup Grant (2020–2021)
Tau Sigma Honors Society